Ratio, Root, and Alternating Series Tests

ANSWER KEY

The Ratio Test

1 Use the Ratio Test to determine whether $\sum_{n=1}^{\infty} \frac{n}{2^n}$ converges.

$\lim_{n\to\infty}\left|\frac{a_{n+1}}{a_n}\right| = \lim_{n\to\infty}\frac{(n+1)/2^{n+1}}{n/2^n} = \lim_{n\to\infty}\frac{n+1}{2n} = \frac{1}{2} < 1$. Converges by the Ratio Test.

2 Use the Ratio Test to determine whether $\sum_{n=1}^{\infty} \frac{n!}{3^n}$ converges.

$\lim_{n\to\infty}\left|\frac{a_{n+1}}{a_n}\right| = \lim_{n\to\infty}\frac{(n+1)!/3^{n+1}}{n!/3^n} = \lim_{n\to\infty}\frac{n+1}{3} = \infty > 1$. Diverges by the Ratio Test.

The Root Test

3 Use the Root Test to determine whether $\sum_{n=1}^{\infty} \left(\frac{n}{2n+1}\right)^n$ converges.

$\lim_{n\to\infty}\sqrt[n]{\left|\left(\frac{n}{2n+1}\right)^n\right|} = \lim_{n\to\infty}\frac{n}{2n+1} = \frac{1}{2} < 1$. Converges by the Root Test.

Alternating series and the Alternating Series Test

4 State the Alternating Series Test, and use it to show $\sum_{n=1}^{\infty} \frac{(-1)^{n+1}}{n}$ converges.

The AST states $\sum(-1)^{n+1}b_n$ converges if $b_n > 0$, b_n is decreasing, and $\lim_{n\to\infty} b_n = 0$. Here $b_n = \frac{1}{n}$: positive, decreasing, and $\to 0$. The series converges (this is the alternating harmonic series).

5 Determine whether $\sum_{n=1}^{\infty} \frac{(-1)^n n}{n+1}$ converges using the Alternating Series Test.

$b_n = \frac{n}{n+1} \to 1 \neq 0$. Since b_n does not approach 0, the Alternating Series Test does not apply, and in fact the series diverges by the nth-term test.

Alternating series error bound

6 Approximate $\sum_{n=1}^{\infty} \frac{(-1)^{n+1}}{n^2}$ using the first 4 terms, and bound the error using the Alternating Series Error Bound.

$S_4 = 1 - \frac{1}{4} + \frac{1}{9} - \frac{1}{16} \approx 1 - 0.25 + 0.111 - 0.0625 = 0.7986$. Error bound: $|S - S_4| \le b_5 = \frac{1}{25} = 0.04$.

Absolute vs. conditional convergence

7 Determine whether $\sum_{n=1}^{\infty} \frac{(-1)^{n+1}}{n^2}$ converges absolutely, conditionally, or diverges.

$\sum\left|\frac{(-1)^{n+1}}{n^2}\right| = \sum\frac{1}{n^2}$, a convergent p-series ($p = 2 > 1$). So the series converges ABSOLUTELY.

8 Determine whether $\sum_{n=1}^{\infty} \frac{(-1)^{n+1}}{n}$ converges absolutely, conditionally, or diverges.

$\sum\left|\frac{(-1)^{n+1}}{n}\right| = \sum\frac{1}{n}$, the harmonic series, which diverges. But the alternating series itself converges (shown earlier by the AST). So it converges CONDITIONALLY.

More Ratio Test practice

9 Use the Ratio Test to determine whether $\sum_{n=1}^{\infty} \frac{3^n}{n!}$ converges.

$\lim_{n\to\infty}\left|\frac{a_{n+1}}{a_n}\right| = \lim_{n\to\infty}\frac{3^{n+1}/(n+1)!}{3^n/n!} = \lim_{n\to\infty}\frac{3}{n+1}=0<1$. Converges by the Ratio Test.

10 Explain why the Ratio Test is INCONCLUSIVE for $\sum_{n=1}^{\infty} \frac{1}{n}$, and identify which test should be used instead.

$\lim_{n\to\infty}\left|\frac{a_{n+1}}{a_n}\right| = \lim_{n\to\infty}\frac{n}{n+1}=1$. Since the limit equals exactly 1, the Ratio Test gives no conclusion. The p-series test (or Integral Test) should be used instead, correctly showing divergence.

Partial sums of an alternating series (visualizing oscillating convergence)

11 Compute the first 5 partial sums of $\sum_{n=1}^{\infty} \frac{(-1)^{n+1}}{n}$ (the alternating harmonic series, which converges to $\ln 2 \approx 0.693$), and describe how they oscillate.

$S_1 = 1$, $S_2 = 1 - \frac{1}{2} = 0.5$, $S_3 = 0.5 + \frac{1}{3} \approx 0.833$, $S_4 \approx 0.833 - 0.25 = 0.583$, $S_5 \approx 0.583 + 0.2 = 0.783$. The partial sums oscillate above and below the true sum $\ln 2 \approx 0.693$, with each successive sum landing closer — a signature behavior of alternating series.

Determining the number of terms needed for a given accuracy

12 For $\sum_{n=1}^{\infty} \frac{(-1)^{n+1}}{n^3}$, find the minimum number of terms needed so the partial sum is within 0.001 of the true sum, using the Alternating Series Error Bound.

Need $b_{n+1} = \frac{1}{(n+1)^3} < 0.001 \Rightarrow (n+1)^3 > 1000 \Rightarrow n+1 > 10 \Rightarrow n \geq 10$. So 10 terms suffice.

Mixed practice: classify each series

13 Classify each series as convergent (state absolutely or conditionally) or divergent:

a $\sum \frac{(-1)^n}{\sqrt{n}}$ Conditionally convergent — converges by AST ($b_n = \frac{1}{\sqrt{n}} \to 0$, decreasing), but $\sum \frac{1}{\sqrt{n}}$ diverges ($p = \frac{1}{2} \leq 1$)

b $\sum \frac{(-1)^n n^2}{n^3+1}$ Conditionally convergent — converges by AST, but $\sum \frac{n^2}{n^3+1}$ diverges by Limit Comparison to $\sum \frac{1}{n}$

c $\sum \frac{(-1)^n}{n!}$ Absolutely convergent — $\sum \frac{1}{n!}$ converges by the Ratio Test (ratio $\to 0$)

The Root Test with more complex expressions

14 Use the Root Test to determine whether $\sum_{n=1}^{\infty} \left(\frac{2n+1}{3n-1}\right)^n$ converges.

$\lim_{n\to\infty}\sqrt[n]{\left(\frac{2n+1}{3n-1}\right)^n} = \lim_{n\to\infty}\frac{2n+1}{3n-1}=\frac{2}{3}<1$. Converges by the Root Test.

15 Use the Root Test to determine whether $\sum_{n=1}^{\infty} \frac{2^n}{n^n}$ converges.

$\lim_{n\to\infty} \sqrt[n]{\frac{2^n}{n^n}} = \lim_{n\to\infty} \frac{2}{n} = 0 < 1$. Converges by the Root Test.

Why the Ratio Test fails for p-series

16 Show that the Ratio Test is inconclusive for EVERY p-series $\sum \frac{1}{n^p}$, regardless of p.

$\lim_{n\to\infty} \left| \frac{1/(n+1)^p}{1/n^p} \right| = \lim_{n\to\infty} \left(\frac{n}{n+1}\right)^p = 1^p = 1$ for ANY value of p. Since the limit always equals exactly 1, the Ratio Test can never determine convergence or divergence for p-series — a different test (the p-series test itself) is required.

Combining multiple tests in one problem

17 Determine whether $\sum_{n=1}^{\infty} \frac{(-2)^n}{n^2}$ converges absolutely, conditionally, or diverges.

Test absolute convergence with the Ratio Test on $\left|\frac{(-2)^n}{n^2}\right| = \frac{2^n}{n^2}$: $\lim_{n\to\infty} \left| \frac{2^{n+1}/(n+1)^2}{2^n/n^2} \right| = \lim_{n\to\infty} \frac{2n^2}{(n+1)^2} = 2 > 1$. Diverges (even in absolute value) — so the original series DIVERGES (the terms don't even approach 0 since $|a_n| \to \infty$).